



Annual Maintenance Completion Report
(2025-2026)
Generating Unit-2



Mechanical Maintenance Unit
Tala Hydropower plant

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1 Overview

The Annual Maintenance of Unit 2 was carried out as per the planned schedule, focusing on key inspection and servicing activities to ensure the continued reliability of the generating unit. Major works included pre-testing and overhauling of nozzles, inspection of the runner, patch weld repaired on runner bucket, replenishment of nitrogen in the MIV accumulator bank, calibration of MIV pressure gauges, commission of the MIV jockey, cleaning of the Main Heat Exchanger, and inspection of the Cooling Water pumps. Additionally, Ultrasonic Testing (UT) was performed on the MIV decompression pipeline, bypass pipeline, and the pressure accumulators of both the MIV and governor systems. All activities were completed in close coordination with relevant sections, maintaining compliance with safety and quality standards.

Scheduled vs Actual date of start and completion of maintenance on the generating unit is shown in Table-1.

1.1 Unit-2 Annual Maintenance

The Annual Maintenance work of the unit commenced on 17th November, 2025 under PTW No: 7289 and completed on 15th December, 2025. The unit was put to mechanical spin at 11:00 hours on 16th December, dry run was initiated at 16:15 hours of 16th December, and shut down at 23: 35 hours upon observing everything normal. The machine was successfully synchronized at 11:50 hours on 19th December.

The major maintenance activities conducted for the unit are detailed hereunder and all test/calibration reports, checklists and inspection notes are placed as Annexure.

Table 1: AM Schedule for Generating Unit II

SN	Unit #	Approved Schedule			Actual Completion		
		Start	End	No. of Days	Start	End	No of Days
1.	Unit II	17.11.2025	28.12.2025	40	17.11.2025	15.12.2025	29

2 Turbine System

Under the Turbine system, the equipment/ component required to be inspected, testing and overhaul includes; Pelton runner, Nozzle injector with deflector Bows, Turbine housing and pit liner.

Table 2: List of Spares Consumed for Turbine System

S. No	Materials	Quantity	
1	M10x60mm long Half threaded Hexa bolt	40	
2	H.T Allen bolt, M20x150 H/thread, Gr.12.9	5	
3	PTFE ring seal 200X209.2X7.5 mm with inner	1	
4	Wiper seal PU with casing110X120X7/10mm,	2	
5	NBR Endless O-ring 240x8mm 70 shore	2	
6	NBR Endless O-ring 350x8mm 70 shore	2	
7	NBR Endless O-ring 270x8mm 70 shore	2	
8	NBR Endless O-ring 330x8mm 70 shore	7	
9	Cup Seal P.P-110X130X12mm	2	
10	NBR Endless O-ring 28x3mm 70 shore	2	
11	NBR Endless O-ring 19.5x3mm 70 shore	1	
12	Wiper seal PU with casing 200X220X12/16mm,	1	
13	Cup seal P.P-200x230x16mm	1	
14	NBR Endless O-ring 246x6mm 70 shore	1	
15	SS stud M12x307 with nut & washer of MCD13	18	
16	SS Allen bolt M10x60, Gr9.8, half thread,	2	
17	M 30x80mm long Half thread socket head bolt	16	
18	H.T Allen bolt M16x160 grade 12.9,	1	
19	SS Allen bolt, M36x340, grade 9.8,	2	
20	NBR Endless O-ring 320x6mm 70 shore	1	
21	S.S Hexagonal Nut & Bolt, M10X65 mm,	5	
22	MS Plain Washer, M8	48	
23	SS Hex. Bolt, M20x180mm A2-80 ISO 3506	2	
24	Hex.bolt-PM16x50-8.8/	4	
25	LED fitting including lamp, 120w	2	
26	Full threaded stud,12.9(HT) M30X1000mm	1	
27	Argon gas 7.00M3/cylinder	21	
28	SS Allen bolt M12x60, Gr9.8 half thread,	6	

2.1 Nozzle Injector Assembly

Table 3: Nozzle particulars

Nozzle Assembly Details	
No. of Nozzles per turbine	5
Material of nozzle body	1.5% Mn steel casting
Material of Needle tip and liner	13% Cr. 4% Ni S.S forging
Material of Jet Deflector	13% Cr. 4% Ni S.S Casting

Net weight of each nozzle	7400kg
System oil pressure	100 Kg/sq.cm
Servomotor type	Double acting, Oil operated with spring closing
Nozzle stroke	210mm
Jet dia.	284
Nozzle Centre line	EL 501.5m

The overhauling and servicing of the nozzle injector is based on oil leakages, water leakages, pre-hydraulic pressure testing and after operational duration of four year without being overhauled. However, no oil or water leakages was observed during its operational hours.

All nozzle injector assemblies underwent functional checks through in-situ oil pressure testing except J₅ because it has exceeded the recommended overhaul interval of four years. During the test, conducted at an initial pressure of 100 bar for a duration of 30 minutes, results were within the acceptable range for remaining four nozzles as shown in table 4.



Figure 1: Nozzle Pre-pressure test

Details of the nozzle pre-test can be found in annexure. An overview of the pre-test result is given in the table below.

Table 4: Details of nozzle pre-test

Nozzle No.	Open/Close Test	Pressure (Bar)	Creeping	Oil Leakage	Date of Testing	Remarks
Nozzle-1	OPEN	Initially decided to dismount considering its recommended overhaul interval of four years.				All the nozzle pre-test are within or close to the acceptable limits. However, nozzle 1 need to be
	CLOSE					
Nozzle-2	OPEN	93	Nil	Nil	18.11.2025	
	CLOSE	83	Nil	Nil	18.11.2025	
Nozzle-3	OPEN	86	Nil	Nil	18.11.20205	

	CLOSE	87	Nil	Nil	18.11.2025	dismounted due to exceeding the recommended overhaul interval of four years.
Nozzle4	OPEN	90	Nil	Nil	20.11.2025	
	CLOSE	78	Nil	Nil	19.11.2025	
Nozzle-5	OPEN	90	Nil	Nil	19.11.2025	
	CLOSE	85	Nil	Nil	19.11.2025	

Therefore, Nozzles 5 was dismantled for overhauling as part of the Annual Maintenance and for the remaining nozzle assemblies that did not require dismantling, were inspected and fasteners were replaced wherever necessary with new set such as such as replacement of needle tip holdings M20x150mm and M10x50mm bolts, O-rings, servicing of deflector shaft bolts. After necessary inspection tag weld was done bolts of nozzle hoods.



Figure 2: Overhauling and servicing of nozzle injector 1



Figure 3: Post-Pressure Test of dismantled nozzle



Figure 4: Changing of needle tip boldings bolts, O-rings, servicing of deflector shaft bolts and tag welding in nozzle boods holding bolts.

2.2 Runner

Table 5: Runner particulars

Particulars	Runner (4020)
Make	Andritz Hydro, Switzerland
Type	Forged Fabricated
Hard Coating	HVOF
Cumulative Running Hours	58,660.37 hours as of 17.11.2025
Repair/ reclamation History	Reclaimed at BHSL during 2018-2019 AM after 29,662.11 running hours

The Unit II runner 4020 has been in operation since its commissioning in 24.05.2012. By November 17, 2025, when the unit was taken offline for the Annual Maintenance 2025-2026, it had accumulated a total of 58,660.37 running hours. The runner was previously reclaimed at BHSL during 2018-2019 AM after completing 29,662.11 running hours. Since the reclamation, it has operated for an additional 28,998.26 hours.

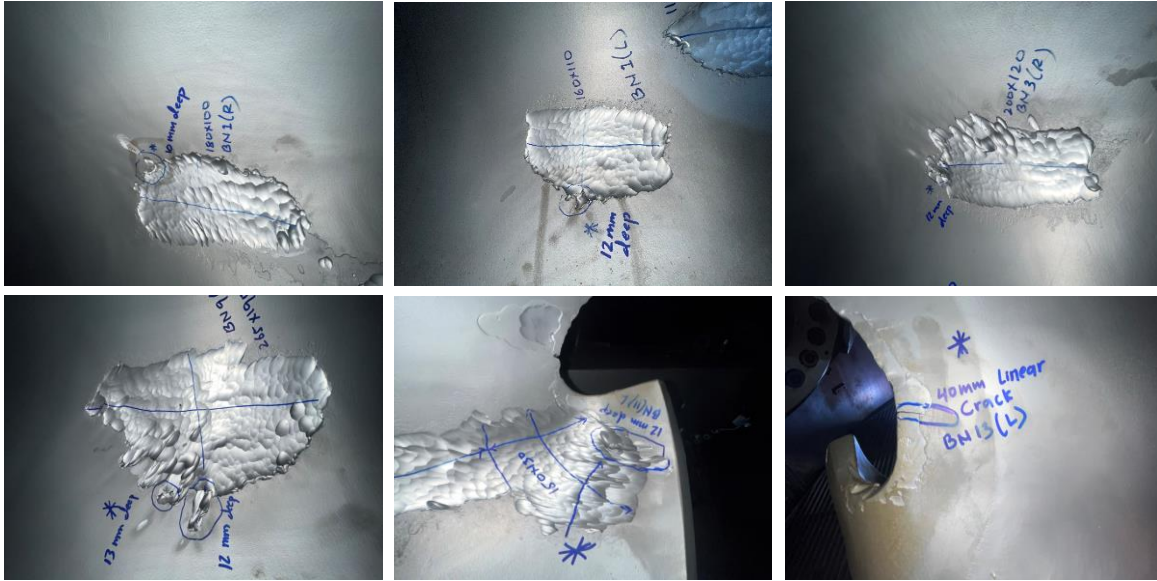


Figure 5: NDT (Visual, DPT AND MPT) Inspection of Micro Gauss Runner 4020

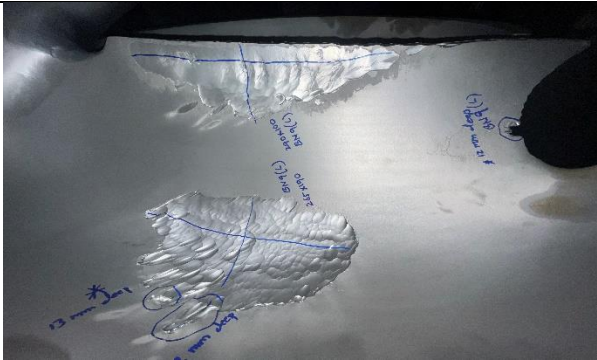
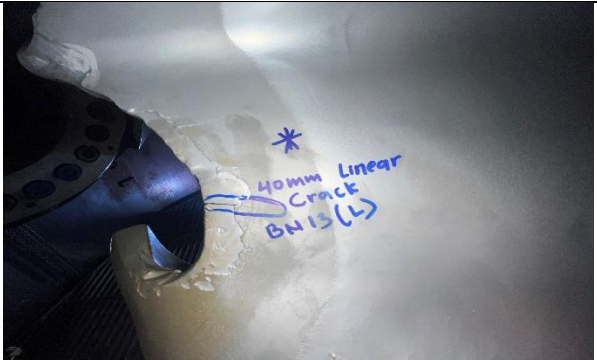
The physical condition of the runner bucket profile is not in good condition because substantial amount of erosion of parent material on every runner bucket is observed besides peeling off of hard coating from almost all bucket surfaces on pressure sides.

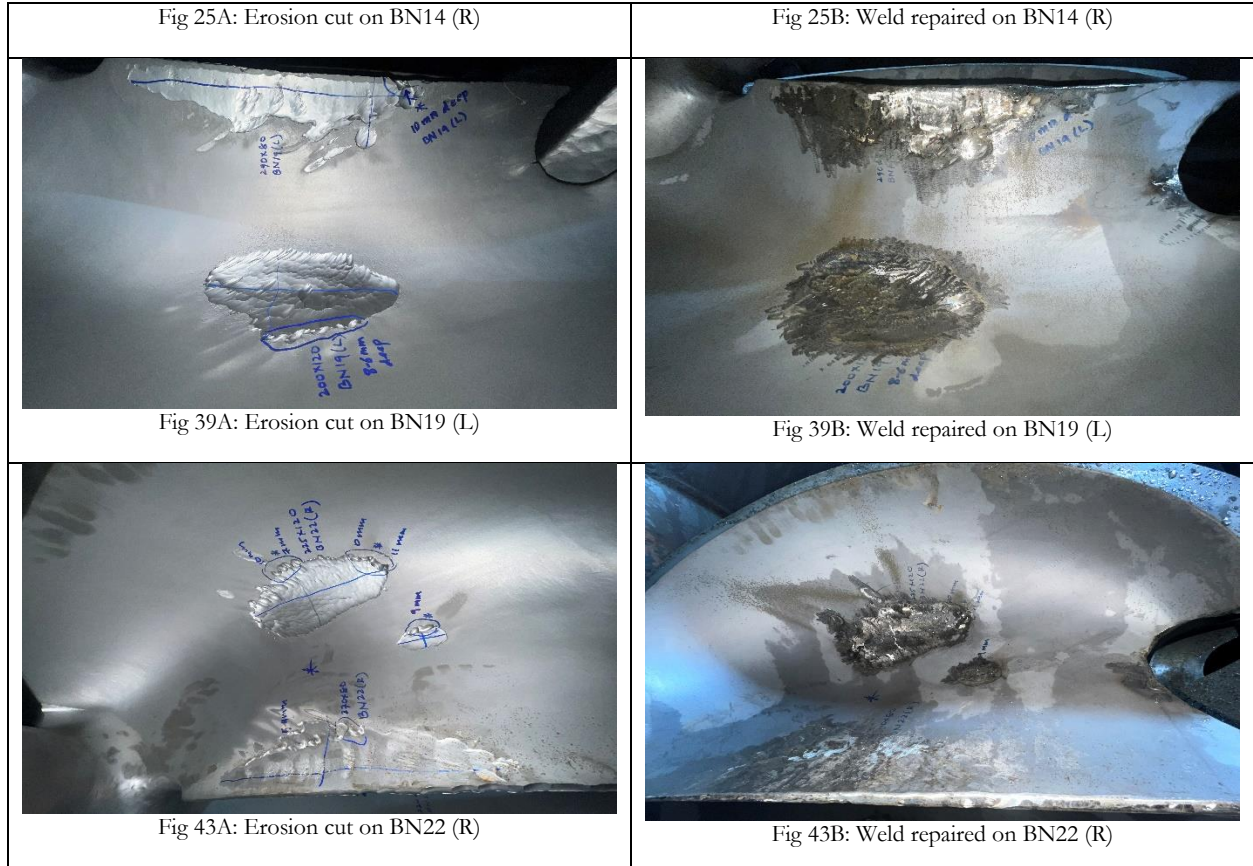
A thorough DNT inspection, including Visual Inspection, Dye Penetrant Test, and Magnetic Particle Test, was carried out to assess the reliability of the runner and determine whether it can sustain one more season. Based on the DNT results, operational hours, and the frequency of weld repairs previously performed on this runner, it has been decided to keep the runner in operation for one more season, provided that necessary grinding and weld repairs are carried out. However, the runner is recommended for replacement in the next annual maintenance.



Figure 6: Runner bucket grinding on eroded part

Table 6: Before and after pictures weld repaired on runner buckets

Before	After
 Fig 15A: Erosion cut on BN9 (L)	 Fig 15B: Weld repaired on BN9 (L)
 Fig 19A: Erosion cut on BN11 (L)	 Fig 19B: Weld repaired on BN11 (L)
 Fig 22A: Liner crack on BN13 (L)	 Fig 22B: Weld repaired on BN13 (L)
 Fig 22A: Liner crack on BN13 (L)	 Fig 22B: Weld repaired on BN13 (L)



The below chart gives information on parent material erosion on each bucket of the runner. The erosion is high on BN; 11 with maximum 21.21% and minimum on BN; 06 with 7.64%. Rest is consistent 10-16% showing the high material erosion on almost all the buckets.

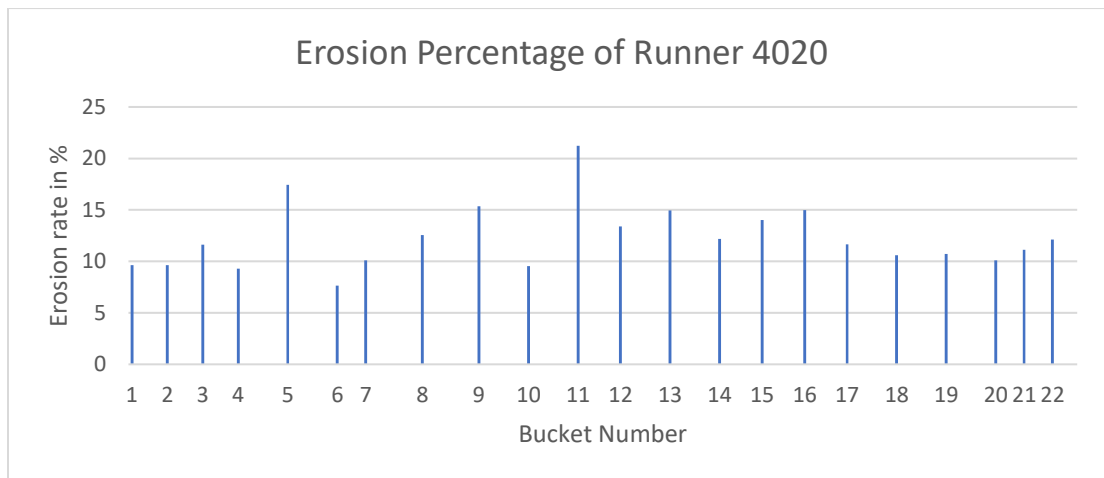


Figure 7: Erosion Area of the Parent Material

2.2.1 Repairing of center frame

The center frame is one of the most critical parts of the underwater components, and its condition must be maintained at all times. Its importance becomes especially evident during the inspection and replacement of other critical underwater components, such as the Pelton runner and nozzle injectors.

During this Annual Maintenance, several broken metal plates were repaired by welding, and additional plates were added wherever there were larger gaps that could lead to mishaps.



Figure 8: Repairing of center frame.

3 Governor and MIV system

Table 7: Spares consumed for Governor and MIV System

S. No	Materials	Quantity
1	Solenoid direction control valve, prt330,	1
2	Adaptor (MS) 3/4" (M)x3/4"(M), rating H	2
3	Seamless SS 3/4"X3/4"NB 90 Degree Elbow	2
4	Seamless SS 1/2"X1/2"90 Degree Elbow	2
5	Seamless SS1/2"X1/2"X1/2"NB Weldable Tee	3
6	Seamless SS1/2"X3/8"Unequal Pipe Adapter	5

7	Seamless SS 3/8"X3/8" Pipe Connector	6
8	3/8" NB Seamless SS pipe connectors	9
9	1/2" NB Seamless SS pipe connectors	16
10	Filter elemt: J/AF-A 0330-R010 BN4HC/-Kb,	2
11	Governor OPU return line duplex filter,	2
12	Filter element-0160 DN 010 BN/HC	2
13	Solenoid valve, 4WE6 D61/WE230 N9Z4L,	1

3.1 Governing System

The main function of governor system is to maintain the rotational speed of turbine within acceptable limits during no-load condition. It performs this function by sensing changes in speed and accordingly controlling flow of water into turbine at fast rate to maintain the speed with reference to grid frequency.

For this unit, the SCADA implementation had been successfully completed during the previous Annual Maintenance. Accordingly, no further works related to the HMC or NCC were required this year, unlike in past maintenance cycles where such activities were a major part of the scope. Instead, routine servicing tasks were carried out to ensure system reliability. These included thorough cleaning of the OPU tank, inspection of pump and oil filters, and replenishment of nitrogen pressure in the accumulator bank to maintain optimal system performance. All high-pressure pipelines associated with VTA (Valve Table Assembly) were carefully inspected for any signs of leakage, and necessary tightening or corrective measures were taken wherever required.

3.1.1 Valve Table Assembly (VTA)



Figure 9: VTA

The VTA is a critical component of the hydraulic control system, primarily responsible for directing and regulating oil flow to the servomotors of the needle and deflector systems. It houses the solenoid valves, pressure gauges, and other control elements that interface with the SCADA system for remote operation and monitoring.

With the SCADA integration completed during the previous Annual Maintenance, the VTA now facilitates more precise and efficient control of the turbine's hydro-mechanical functions, reducing the need for manual intervention and improving overall system responsiveness.

The functionality test of the system was checked and necessary setting was carried out. General cleaning and servicing of Idler valve assembly was carried out. The OPU sump was oil-drained and cleaned, the O-rings were replaced and the filters were replaced with new ones.

The N₂ gas piston accumulator for Governor OPU was checked and its functionality confirmed. Replenishment of N₂ gas to the required pressure and replacement were done as per the condition of the return line duplex oil filter and filter elements.

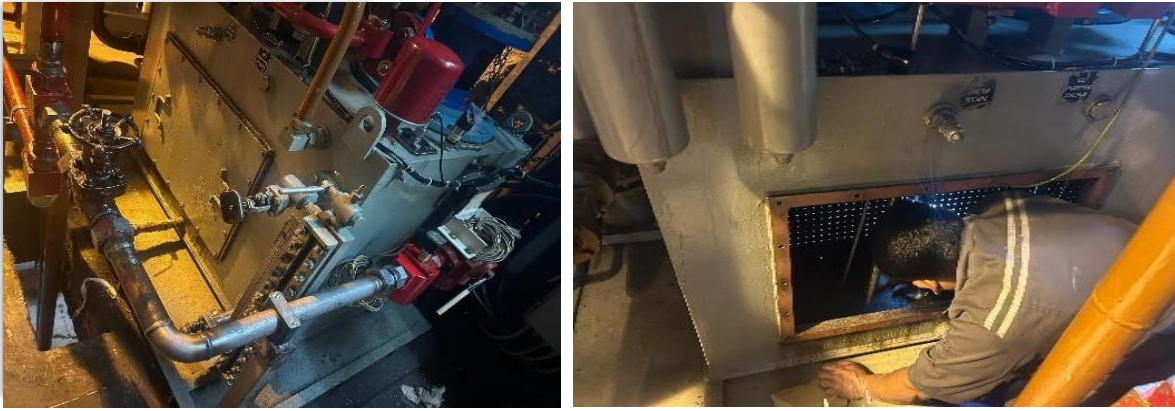


Figure 10: Servicing of Governor OPU

Deflector gap setting was done and the set result is tabulated below.

Table 8: Deflector gap setting

Nozzle	Initial reading (mm)	Final reading (mm)
J1	6	6
J2	9	8
J3	11	8
J4	19	8
J5	16	8

3.2 MIV System

The MIV system is the primary isolation for turbine and principal protection for the station against penstock water. The N₂ gas bank for MIV was replenished and tested for N₂ gas leakage with surf foam bubble.



Figure 11: Leakage checking of N2 gas bank for MIV

The pressure gauges of MIV control panel and accumulator have been calibrated at HRDC and all are found to be in good working condition. The report is attached in the annexure.



Figure 12: Removing of MIV pressure gauges to be sent for Calibration

In order to ensure safe parameters of the control pipelines for safe and smooth operations, periodic NDT is highly recommended since these are considered among the most critical parts.

The ultrasonic testing of MIV bypass pipeline, decompression pipeline and pressure accumulators for thickness verification will be done later on along with other 2 units by an inspection team from HRDC to inspect any possible deviations in the pipeline thickness or any crack development before it amplifies. We are unable to conduct UT because of time constrain and unavailability of inspection equipment and inspectors at our required time.

OPU pump capacity was checked and found normal. Functionality check for internal leakages and servicing was carried on all the directional control solenoid valves provided in OPU and MIV control panel. Contactors and limit switches of MIV were checked for their healthiness and replacement done wherever necessary. During the functionality check, no N2 gas leakage was observed.

4 Cooling Water System

Table 9: Spares consumed for Cooling Water system

S. No	Description	Quantity
1	Wearing ring CF8M, part# 190.0	2
2	NBR Endless O-ring 78x4mm 70 shore	4
3	H.T hex. head, bolt & nut - M6 x 20 mm	14
4	Line shaft bearing-part no. 33, Br	1
5	S.S ball valve -3/8"	3

4.1 Clean Cooling Water System

During this Annual Maintenance, CCW pumps B underwent a complete inspection with dismantling, cleaning, and overhauling of critical components such as the mechanical seals, bearings, impellers, and shaft sleeves. Worn-out parts were replaced wherever necessary.

Following the overhauling, servicing and reassembly, shaft alignment between the pump and motor was carried out using an Easy-Laser alignment system. The laser alignment process ensured accurate correction of both angular and parallel misalignments, thereby minimizing vibration and mechanical load on the coupling. Final alignment readings were recorded and confirmed to be within acceptable tolerance limits. The alignment readings have been attached as annexure.



Figure 13: CCW pump alignment check

4.2 Raw Water-Cooling System

Two vertical turbine pumps one as main and other as standby are provided for cooling of clean water at Heat Exchangers, Generator Transformer and shunt Reactor. Both the main and standby RCW pumps were inspected for mechanical integrity and operational readiness and found to be healthy.

Motorized water strainer was overhauled and cleaned thoroughly, replaced digital flow meter and assembled back. The debris accumulated on strainer cones were huge and is recommended to clean during every annual maintenance during prevent the debris accumulation and choking of outlet holes. The NRVs of both the RCW pumps of the unit were inspected and found to be healthy.

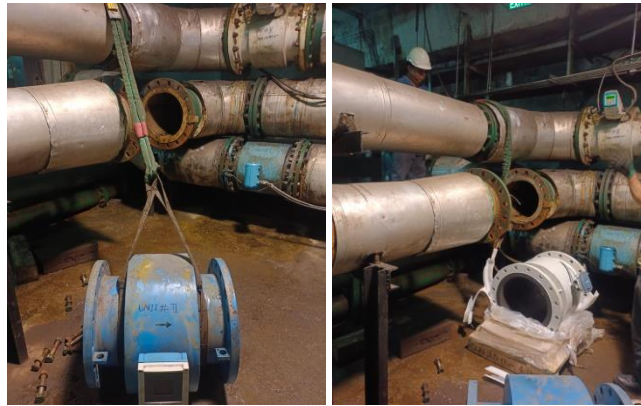


Figure 14: Replacement of digital flow meter.



Figure 15: Cleaning and servicing of RCW strainer.

4.3 Main Heat Exchanger (MHE)

Each unit is equipped with a shell-and-tube type heat exchanger (MHE) to facilitate the circulation of clean cooling water for the stator air coolers, OPU coolers, and bearing coolers. Raw cooling water is supplied through the tubes inside the heat exchanger to enable heat transfer.

During the annual maintenance, the heat exchanger was opened, and the tubes were thoroughly cleaned and flushed to remove any scaling or deposits that could impede heat transfer efficiency. After cleaning, the end covers were reassembled, and pressure testing was conducted at 8kg for 30 minutes to ensure there were no leaks and that the unit was fit for continued operation.



Figure 16: Cleaning and servicing of Heat Exchanger.

4.4 Cyclone Separator

The cyclone separator is used to trap the coarse sediments from the raw water for unit was also overhauled for cleaning and servicing. The sediments collected at the bottom during operations are taken out by hand and flushing method.



Figure 17: Flushing of sediments from cyclone separator

Internal/ external services availed

- Runner inspection (Magnetic Particle Inspection) by HRDC
- Ultrasonic testing on decompression, bypass pipelines, pressure accumulators and distributor pipeline joints will be conducted by HRDC
- Vibration analysis of CCW and RCW pumps by HRDC
- Calibration of pressure gauges on MIV panels by HRDC
- Testing of Governor and MIV system oil samples by HRDC

5 Maintenance Challenges

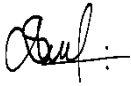
The overhauling of the turbine system involved substantial effort, particularly in dismounting and overhauling three nozzles. This task was physically demanding and required careful handling, but it was executed smoothly with effective teamwork and coordination.

Annexure

S.NO	Annexure	Description
1	Annexure 1	Unit 2 PTW
2	Annexure 2	Nozzle pre-testing
3	Annexure 3	Underwater component inspection sheet
4	Annexure 4	Unit 2 Runner MPT Report.
5	Annexure 5	Nozzle Post Assembly Checklist
6	Annexure 6	Nozzle Assembly Checklist
7	Annexure 7	Fasteners Tightness Checklist
8	Annexure 8	Turbine final check list
9	Annexure 9	Governor checklist cum OK card
10	Annexure 10	Gov OPU pump capacity check
11	Annexure 11	Gov OPU safety valve setting
12	Annexure 12	Gov N2 pressure retention check
13	Annexure 13	Gov system pressure retention check
14	Annexure 14	Deflector servo timing set
15	Annexure 15	Deflector mechanical gap setting
16	Annexure 16	G100 Governor system functionality check
17	Annexure 17	Nozzle opening closing timing check
18	Annexure 18	MIV checklist cum OK card
19	Annexure 19	MIV OPU safety valve setting
20	Annexure 20	MIV pressure retention check

21	Annexure 21	Inspection of Decompression Servo Valve
22	Annexure 22	MIV system timing set
23	Annexure 23	Governor final checklist
24	Annexure 24	Auxiliary Pre-Checklist
25	Annexure 25	Final Checklist of Cooling water system
26	Annexure 26	Main Heat Exchanger Pressure Test and Leakage Test.
27	Annexure 27	Auxiliary final checklist
28	Annexure 28	Pressure Gauges Calibration Report

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